INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Spring Semester Examination 2023-24

Date of Examination: ________________ **Session: (FN/AN)** _____ **Duration: 2 hrs.**

Full Marks : 30 Subject No.: NA20202 **Subject : Marine Hydrodynamics**

Department/Center/School: Ocean Engineering and Naval Architecture

Specific charts, graph paper, log book etc., required

Special Instructions (if any): NO

Please note that the notations and variables used in the question paper are same as that of used in the class and hence not explained here.

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1. Derive the Continuity equation for Cartesian co-ordinate system. 3

2. Write the equation of motion for a fluid which is incompressible, homogeneous, inviscid in nature in Cartesian co-ordinate system. Further modify the equation if the flow is considered to be ir-rotational. Discuss the importance of the linear dynamic term in the pressure equation over the quadratic term in the context of marine hydrodynamics. 5

3. If the velocity potential is given by the equation $\phi(x, y) = xy$, what is the stream function? Also write the equation of the stream line, and find the steady pressure. 3

4. In which condition, $\phi = xf(r)$ becomes a fluid flow? 4

5. If the radial velocity of a flow is described by the equation $V_r = \frac{k}{\sqrt{r}} cos\theta$. if $V_\theta = 0$ at $\theta = 0$, Find the rotational velocity and the stream function. 5

6. A source at origin and a uniform flow at 5m/s is superimposed. The half body which is formed has a maximum width of 2m. Calculate the following:

a) Stagnation point

b) width of the half body at the origin.

c) velocity at the point (0.7, $\pi/4$) 5

Or

7. Write down the complex potential of a flow which is superimposing a uniform flow and a rotating cylinder of a given radius. Also find the stream line of that flow.

8. Show that $G(x,y,z) = \frac{1}{r}$ where $r^2 = x^2 + y^2 + z^2$, satisfies the Laplace equation everywhere except the origin and from that results, show that velocity potential $\phi(P)$ at any point P in an unbounded fluid can be given as $\phi(P) = -\frac{1}{4\pi}\left(\iint_S \left(\phi(Q)\frac{\partial}{\partial n}\left(\frac{1}{r}\right) - \frac{1}{r}\frac{\partial \phi(Q)}{\partial n}\right) dS\right)$. 5